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"""Part C-1:GUI program to design a Simple Calculator"""

from tkinter import *
win=Tk()
win.geometry("312x324")
win.resizable(0,0)
win.title("Calculator")

def btn_click(item):
    global exp
    exp=exp+str(item)
    itxt.set(exp)

def btn_clear():
    global exp
    exp=""
    itxt.set("")

def btn_equal():
    global exp
    result=str(eval(exp))
    itxt.set(result)
    exp=result

exp=""
itxt=StringVar()
#first frame
iframe=Frame(win,width=312,height=50,bd=0,highlightbackground="black",
highlightthickness=2)
iframe.pack(side=TOP)
ifield=Entry(iframe,font=('arial',18,'bold'),textvariable=itxt,width=50,
bg="grey",bd=0,justify=RIGHT)
ifield.grid(row=0,column=0)
ifield.pack(ipady=10)

#button frame
btframe=Frame(win,width=312,height=272.5,bg="grey")
btframe.pack()

#zero row
clear=Button(btframe,text="Clear",fg="black",width=32,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_clear()).grid(row=0,column=0,columnspan=3,
padx=1,pady=1)
divide=Button(btframe,text="/",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click("/")).grid(row=0,column=3,padx=1,
pady=1)

#first row

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seven=Button(btframe,text="7",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(7)).grid(row=1,column=0,padx=1,pady=1)
eight=Button(btframe,text="8",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(8)).grid(row=1,column=1,padx=1,pady=1)
nine=Button(btframe,text="9",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(9)).grid(row=1,column=2,padx=1,pady=1)
multi=Button(btframe,text="*",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click("*")).grid(row=1,column=3,padx=1,pady=
1)
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#second

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four=Button(btframe,text="4",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(4)).grid(row=2,column=0,padx=1,pady=1)
five=Button(btframe,text="5",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(5)).grid(row=2,column=1,padx=1,pady=1)
six=Button(btframe,text="6",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(6)).grid(row=2,column=2,padx=1,pady=1)
minus=Button(btframe,text="-",fg="black", width=10,height=3, bd=0,
bg="white",cursor="hand2",command=lambda:btn_click("-")).grid(row=2,column=3,
padx=1,pady=1)
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#third

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one=Button(btframe,text="1",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(1)).grid(row=3,column=0,padx=1,pady=1)
two=Button(btframe,text="2",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(2)).grid(row=3,column=1,padx=1,pady=1)
three=Button(btframe,text="3",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(3)).grid(row=3,column=2,padx=1,pady=1)
plus=Button(btframe,text="+",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click("+")).grid(row=3,column=3,padx=1,
pady=1)
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#fourth

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zero=Button(btframe,text="0",fg="black",width=21,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(0)).grid(row=4,column=0,columnspan=2,
padx=1,pady=1)
dot=Button(btframe,text=".",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_click(".")).grid(row=4,column=2,padx=1,
pady=1)
eq=Button(btframe,text="=",fg="black",width=10,height=3,bd=0,bg="white",
cursor="hand2",command=lambda:btn_equal()).grid(row=4,column=3,padx=1,pady=1)
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win.mainloop()
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